

Artificial Intelligence & Systems Laboratory

In collaboration with TIM & AMS Laboratories

Organize the :

International Conference on Artificial Intelligence & Systems ICAIS'25

**Main Theme: «Bridging Theory and Applications for Innovation,
Sustainability and Societal Challenges»**

27 to 28 November 2025 at the Faculty of Science Casablanca

Conference Program

Day 1: Thursday, November 27, 2025

8H30-9H00

Welcome and registration of participants

9H30-10H00

Welcome and Opening Speech

- President of Hassan II University of Casablanca
- Dean of Faculty of Sciences Ben M'Sick
- Vice deans of Faculty of Sciences Ben M'Sick
- Head of the Mathematics and Computer Science Department
- Director of TIM laboratory
- Director of AMS laboratory
- Director of IAS laboratory
- Conference Chair : Pr. A. BELANGOUR

Moderator : Pr. M. AIT DAOUD

Plenary session: Amphi 8

10H00-11H00

Keynote 1: Cognitive AI for Inclusive Health: Towards Intelligent Systems Supporting Maternal and Child Care in Low-Resource Contexts

Speaker: Pr. Khadija ACHTAICH

State Engineer and researcher in the field of Artificial Intelligence. Professor at the Faculty of Sciences Ben M'Sick (FSBM), Hassan II University of Casablanca in Morocco.

Moderator: Pr. A. MARZAK



Faculty of Sciences Ben M'Sick

11H00-12H00 **Keynote 2: Artificial Intelligence for Health and Wellbeing**
Link: https://meet.jit.si/ICIAS_Keynote2

Speaker: Pr. Anderson ROCHA
Full-Professor at University of Campinas, Brazil.

Moderator: Pr. E. BENLAHMAR

12H00-12h30	Coffee Break and Networking
12H30-14H	Parallel Sessions (Block 1)

Session 1: AI for Software Engineering & Cybersecurity

Chairs : Pr. M. AZZOUAZI, Pr. M. HAIN, Pr. F. ELAAZABY, Pr. S. OUAHABI, Pr. K. BOUSKOURA

K. LETRACH, K. EL BOUCHTI, N. AMIMAR, S. ZITI: Secure and Intelligent Software Engineering: Automatic Detection and Correction of Code Vulnerabilities Using Generative AI Agents
N. EL INANI, F. BENABBOU, S. OUAHABI, K. SABIRI: Enhancing Smart Contract Security through Transformer-Based AI Models
S. BAYAT, M. RAHMOUNI, S. ZITI: Intelligent Optimization of Software Test Automation to Enhance Efficiency and Software Quality
N. EL-KEBIR, Y. ZAOUI, D. RIANE, S. ZITI: A Graph-Theoretical Approach to Intelligent Threat Detection in Financial Cloud Systems
M. CHAOUI, S. ZITI: Machine Learning applied to Malware as a Service Driven attacks detection

Session 2: NLP for Arabic & Low-Resource Languages

Chairs : Pr. M. HANOUNE, Pr. L. MOUSSAID, Pr. M. IZID, Pr. K. KANDALI, Pr. H. OUCHRA

A. SEBBAR, O. ZAHOUR, E. BENLAHMAR, F.I MOUNTASSIR, T. AMZIL: Sentiment Analysis for Arabic Dialects: An Experimental Study
Y. MATRANE, Z. ELLAKY, F. BENABBOU, M. AIT DAOUD: Toward a Conditional GAN-Based Framework for Class Imbalance Mitigation in Arabic Dialect
S. TACE: Enhancing Arabic Language Understanding: AraBERT Models for Frozen Expression Identification
E. HAIDARA, Y. SEKHARA, A. BELANGOUR: A Multicriteria Comparative Framework for Adapting Arabic Large Language Models to Uşul al-Fiqh Reasoning

Session 3: AI-Powered Personalized Education



Chairs : Pr. A. MARZAK, Pr. M. MOUTACHAOUK, Pr. I. BENAMRI, Pr. T. AHAJJAM, Pr. M. BACHRAOUI

S. SAMMAH, M. AIT DAOUD, K. ACHTAICH: From Data to Decision: A Conceptual Framework for AI-Powered Student Guidance in Higher Education

R. BACH, K. ACHTAICH, H. OMARI: Intelligent Chatbot for Mathematics Learning Assistance

A. QEQLIN : Détection et prévention de l'abandon scolaire grâce à l'intelligence artificielle multimodale

A. ECHCHOAYEBY, W. ABBAOUI, S. ZITI: AI-Based Modeling and Optimization for Fair and Explainable Personalization of Educational Pathways

H. EZZAHR, ZAOUI SEGHRUCHENI, K. SALAH-EDDINE: The Multimodal AI as a Solution to Higher-Learning-Accessibility

Session 4 : AI for Public Governance & Smart Cities

Chairs : Pr. E. BENLAHMAR, Pr. R. AITABDELOUAHID, Pr. N. AHARRANE, Pr. Y. CHEMLAL, Pr. H. FERJOUCHIA

C. ERRACHID, M. AIT DAOUD, K. ACHTAICH: Towards a Smart University in Morocco: The Role of Artificial Intelligence in Optimizing Governance Processes

L. MOULAD, C. WATIK : La collaboration humain-IA au service de la gouvernance intelligente : vers un modèle intégré de développement d'une ville intelligente – cas du Maroc.

Y. AMZOU, S. HAMIDA, S. ZITI: Agentic AI for MLOps/DataOps Automation: Towards Predictive and Distributed Decision Systems for Smart National Governance.

N. MAMDOUH : Souveraineté numérique et transition énergétique : Vers un nouveau paradigme de résilience pour les organisations marocaines.

I. TAYMI, O. AITLMOUDDEN, N. AHARRANA, M. AITDAOUD: Agro-systemic modeling and AI for visual perception and optimization of production processes.

14H-15H	Lunch Break
15H-16H30	Posters Sessions

Chairs: Pr. T. AHAJJAM, Pr. H. BOUTAYEB, Dr. O. AITLMOUDDEN, Dr. Y. MATRANE

N. BELAATIK, F. BENABBOU : Capacity-Constrained Clustering for Data-Driven Zoning of Informal Retail Networks in Morocco

O. EL OUADH, M. GHAZOUANI: Real Time Traffic Sign Detection and Classification Using Computer Vision

A. BAQUOUCH, N. SAEL, A. LAAMOURI : Exploring the Potential of Multimodal Large Language Models for Advanced Recommendation Systems

Z. DRISSI, S. OUNACER: Adaptive Virtual Tutoring System Using Large Language Models and Bayesian Knowledge Tracing

H. BOUTAYEB: Automatic Bias Detection in Video Interviews Using AI and Linguistic Analysis



- S. ETTOUSSI:** Adaptative Tool for Education on Moodle
- A. NADIR :** Segmentation des lésions cérébrale par imagerie médical
- A. ZOUBIR, Y. SEKHARA:** Artificial Intelligence for the Automatic Evaluation and Correction of Arabic Phonetic Rules
- I. BOUBOUL, R. AIT ABDELOUAHID:** RAG-based Intelligent Agent for Automated Retrieval and Document-Level Summarization of Scientific Publications
- H. TIJANI, Y. SEKHARA :** Détection d'anomalies dans les connexions réseau internes par apprentissage automatique
- N. ELALAMI, F. BENABBOU:** Improving information extraction from invoices by enriching OCR results with pre-trained language models (LLMs)
- A. ELGOUJI :** Détection et classification des panneaux de signalisation routière
- I. FARKHANE:** Intelligent Football Player Performance Analysis System Using Computer Vision and Deep Learning
- H. BARHRHOJ:** LLM-Assisted Automatic Educational Content Generation Platform
- N. EL HOUMA :** Exploitation du Process Mining pour l'analyse des commandes e-commerce et la détection des anomalies et fraudes
- S. BEN MANSOUR, M. SADIQ:** AI-Driven Legal Intelligence: A Retrieval-Augmented System for Semantic Search and Interpretation of the Moroccan Family Code.
- A. MOHAMMED, R. AIT ABDELOUAHID:** Multimodal AI Agent for Automated Document Management and Analysis: Translation, Summarization and Tone Personalization
- M. KHALIL ABOUSSAAD, R. ABDELOUAHID :** IA Multimodal pour la Recherche Intelligente des Données à partir de Tableaux via LLM et Embeddings.
- Z. AKDIM, M. AIT DAOUD:** Autonomous customer support agent
- M. ALLA, M. AIT DAOUD:** Automatic Real-Time Credit Card Fraud Prevention Using Artificial Intelligence
- N. EL MANSOUR, A. BELANGOUR :** Multi-Purpose Banking AI Assistant: Integrating LLM and RAG in a Web-Based Platform with Back-Office Analytics
- A. GANDAFFA, M. SADIQ:** Multimodal AI System for Automated Video Interview Analysis and Resume Job Compatibility Assessment
- O. RIAFI :** Agent IA Multimodal pour l'Analyse de Données Structurées via Chatbot et Bases de Données.
- S. ILYASS, Y. SEKHARA:** An AI-Powered Cognitive Behavioral Therapy Chatbot with Crisis Detection and Personalized Therapeutic Interventions
- W. ZAYD, F. BENABBOU, Z. ELLAKY, Y. MATRANE:** Cyberbullying Activity Detection over Social Media using MultiModal LLMs

15H-17H00

Parallel Online Sessions (Block 1)

Session 1 : https://meet.jit.si/ICAIS25_session1

Chairs : Pr. A. ETTOUFIK, Pr. Z. ELLAKY, Pr. S. BOUHSSISIN, Dr. Y. MATRANE, Dr. M. LABIED

M. BENAKRIM, A. ETTOUFIK, A. MAIZATE: Enhancing Resilience of Energy CPS Through Deep Learning-Based Anomaly Detection



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I. EL HARNAF, K. ACHTAICH, S. TETOUANI: Fault Type Prediction in Industrial Systems Using a Hybrid TCN–XGBoost Framework for Predictive Maintenance
N.I MAMDOUH, M. HAJJAJ, E. RAHOU : Souveraineté Numérique et Transition Énergétique : Vers un Nouveau Paradigme de Résilience pour les Organisations Marocaines
R. ILAL, Y. ZAOUI SEGHROUCHENI, S. ZITI: Multimodal AI for the Detection and Prevention of School Dropout

Session 2 : https://meet.jit.si/ICAIS25_session2

Chairs : R. AITABDELLOAHAD, Pr. H. OUCHRA, Pr. K. KANDALI, Dr. N. SAFEH

F. ABNOULGID, S. AOUHASSI, K. MANSOURI, F. AKEF : L'Université 4.0, un Levier pour l'Innovation Durable : Un Cadre Pédagogique Propulsé par l'IA pour le Développement de Compétences en Résolution de Problèmes Complexes liés aux ODD
M. ZOKAY, H. SAYLANI: Skin diseases classification using artificial intelligence and exploiting new dermatological images representations
M. S. SIDI ELY: Understanding Blockchain and Its Impact on Data
S. ABDELLAOUI, W. ABBAOUI, S. ZITI: A Narrative Study of SQL to NoSQL Database with Perspectives on an AI-Based Migration Approach
M. SENNAD, F. BENABBOU, Z. ELLAKY, R. AITABDELOUAHID: Transformer Based Classification of Dark Web Market Activities
I. CHABBAKI, S. EL FILALI, S. OUAHABI: Sentiment Analysis in Moroccan Darija: Hybrid Annotation and Comparative Evaluation of Machine Learning, Deep Learning, and Transformer Models
S. YAMARI, E. BENLAHMAR: VAE-GUARD: An Edge-Load-Balancer Framework Combining Sequential VAE Anomaly Detection with HA-Proxy

15H-17H00

Parallel Sessions (Block 2)

Session 5 : Computer Vision for Detection & Monitoring

Chairs : Pr. N. SAEL, M. GHAZOUANI, Pr. Y. SEKHARA, Pr. A. ZAOUCH, Pr. H. BOUTAYB, Pr. S. ZITI

A. BAALA, M. HANOUNE, M. BENTAIB: Fine-Tuned YOLOv8n Model for Enhanced Violence Detection in Surveillance Systems.
S. MOUANID, F. BENABBOU: Automated system for tomato leaf detection and classification
H. OUCHRA, A. BELANGOUR : Towards intelligent mapping of urban areas: application of the YOLO v11 model to building detection
H. ASSEMLALI: Real-Time Object Detection and Monocular Distance Estimation for Autonomous Vehicles
M. EN-NAJIDI, K. ACHTAICH, H. OUCHRA: AI Meets Safety: A Vision-Based Approach for PPE Monitoring and Risk Prevention
A. SBAYJI, A. OUACHA, S. ZITI: Automatic Detection of Non-Compliant Constructions Using Artificial Intelligence from Multi-Source BIM and GIS Data
M. OULMA, K. EL BOUCHTI, S. ZITI: Deep Learning-Based Intelligent Framework for Urban Anomaly Detection and Smart City Monitoring



O. GUENDOUL , Y. TABII, R. OULAD HAJ THAMI: Video-Based Action Recognition for Elderly Care Using Skeleton Data
T. BOUCHAMA, A. ETTOUFIK, Y. SEKHARA : Détection intelligente des objets dynamiques dans les scènes vidéo en temps réel par la vision par ordinateur et l'IA: Application Smart City
O. AITLMOUDDEN, I. TAYMI, A. NKAILI, L. MOULAD, M. AITDAOUD: Using Drones in Treating Plantations: An Agricultural Engineering Perspective
S. BOUHSSISSIN, N. SAEL: Performance Analysis of Multimodal AI Models for Structured Data Extraction

Session 6 : AI Tools for Learning & Engagement Analysis

Chairs : Pr. M. SADIK, Pr. M. DAIF, Pr. S. BOUHSSISSIN, Pr. S. BENKADDOUR, Pr. EL ADRAOUI

F.Z. OUAHOUDA, K. ACHTAICH, N. ACHTAICH: Teaching Artificial Intelligence with Scratch in Primary Education: Challenges and Prospects
H. OMARI, K. ACHTAICH, K. HATTAF: Mathematics Learning Through an AI Platform for Middle School Students
S. CHAIBI, M. RADID, G. CHEMSI: Impact of Using the Scientific Inquiry Approach with Computer Simulation on Students' Understanding of Fundamental Physics Concepts
N. SAFEH, O. AIT LMOUDDEN, S. OUAHABI: Emotional Dashboard: Analysis of Student Emotions in Online E-Learning Quizzes
I. QARBAL, H. ABOURIFA, N. SAEL, S. OUHABI: Privacy-Preserving Student Engagement Detection Using Temporal Deep Learning
Z. RAMZI, K. ACHTAICH, M. AIT DAOUD: Artificial Intelligence and Process Mining Approaches for Adaptive and Personalized Learning in Immersive Contexts
F. ABBOUR, S. ARDCHIR, S. OUNACER, M. AZZOUAZI: Advancing Fairness and Equity in AI-Driven Recruitment: Unveiling Hidden Patterns of Intersectional Bias
C. ZAOUI, C. BOUAINE, F. BENABBOU: Intrusion Detection Using Machine Learning Techniques for Data Analysis.
A. KHADDAR, Y. SAID, B. BELLA, A. DEHBI, T. CHAFIQ: Artificial Intelligence, IoT, and Big Data in Education: Opportunities and Challenges for Smart Learning Environments
Y. AMZOU, S. HAMIDA, S. ZITI: Customize learning paths with AI integration to help enhance learning.

Session 7: Multimodal Artificial Intelligence for Innovation in Education, Plant Health, and Advanced Data Analysis

Chairs : Pr. SAEL, Pr. K. ACHTAICH, Pr. AIT DAOUD, Pr. KANDALY, Pr. I. BENAMRI

Z. EL Fakir, O. KAICH, O. ZAHOUR, S. EL FILALI, E. BENLAHMAR: Simple pre-ASR noise filtering to improve robustness of speech-to-text (STT) in low-resource, code switching settings.
O. KAISSOUNI, S. OUAHABI, N. SAEL: Multimodal and Multilingual Deep Learning Approaches for Automated Plant Disease Diagnosis: A State-of-the-Art Review
M. BARCHANE, E. BENLAHMAR, O. ZAHOUR: **Beyond Counts:** A Semantic Metric for Scientific Influence
I. TALKAM, A. BELANGOUR, I. HAMZANE: Towards a Probabilistic and Transparent AI System for



Agile Project Cost Estimation

B. EL ALEM, C. MOUJAHID, S. ZITI: Enhancing Educational Inclusion through an Interactive AI-based Sign Language Translation System

M. AMRANI, Y. ZAOUI SEGHRUCHENI, K. SALAH-EDDINE: Enhancing Accessibility in Education: A Personalized and Multimodal Learning Approach for the Deaf Community

B. BELLA, Y. CHEMLAL, M. AZZOUAZI, S. OUNACER, A. KHADDAR: The Influence of Psychological Factors on Graduates' Employability: A Comprehensive Review of Data Mining Approaches

F. AIT DAOUD, A. BABAKHOUEA, S. YANISSE: Artificial Intelligence Applications for Optimized Management of Medical Devices

17H00-17H30

Coffee Break

Day 2: Friday, November 28, 2025

Plenary session: Amphi 8

9H00-10H00

Keynote 3 : Data, Optimization and Machine Learning

Link: https://meet.jit.si/ICIAS_Keynote3

Speaker: Pr. Ana Isabel PEREIRA

Vice-coordinator the Research Centre in Digitalization and Intelligent Robotics (CeDRI) - Polytechnic Institute of Bragança, Portugal.

Moderator: Pr. Sanaa ELFILALI

10H00-11H00

Keynote 4 : L'Intelligence Artificielle au Service de la Santé

Speaker: Mr. Radouane MASTOURI

Founder and CEO of Precision Systems, a startup focused on technological innovation in healthcare. He leads the development of Doctify Health, AI-powered Electronic Health Record (EHR), Morocco.

Moderator: Pr. Nawal SAEL

11H00-11h30

Coffee Break and Networking

11H30-13H00

Parallel Sessions (Block 3)

Session 8 : Advanced AI in Medical Imaging & Specialized Vision

Chairs : Pr. M. BENTAIB, Pr. M. AIT DAOUD, Pr. S. AREZKI, Pr. A. ZAKRANI

C. EL KHASSOUMI: Revolutionizing Medical Imaging with Artificial Intelligence

C. AL ALOUANI: Deep Learning in Psychiatric Facial Recognition: A Comparative Synthesis of 2D and 3D Approaches

H. ABOURIFA, I. QARBAL, S. OUAHABI, N. SAEL, K. EL GUEMMAT: Detecting Cheating Behaviors During Exams via Emotional Analysis



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O. KAICH, Z. EL FAKIR, S. EL FILALI, O. ZAHOUR, E. BENLAHMAR: Expert-in-the-Loop AI Framework for Authentic 3D Reconstruction of Historical Scenes

Session 9 : Large Language Models and Intelligent Dialogue Systems

Chairs : Pr. F. BENABBOU, Pr. M. EDDAOUI, Pr. K. ELGUEMMAT, Pr. I. AGMOUR, Pr. I. EL BERRAI

O. AITLMOUDDEN, O. NAYA, N. SAEL, S. BOUHSISSIN: Design and Implementation of a RAG Chatbot for Moroccan Accounting and Taxation

Z. ELLAKY, Y. MATRANE, F. BENABBOU, A. ZAOUCH, K. ACHTAICH, R. AIT ABDELOUAHID, A. SABOR, Y. FARIH: Automatic Detection of Online Hate Speech Using Transformer-Based Models

M. KASBOUYA, N. LOUZI, N. SAEL: Multi-Domain Training for Emotion Recognition in Dialogue Systems

T. AMZIL, A. JANNANI, R. AIT ABDELOUAHID, N. SAEL : Topic Modeling for Moroccan Darija Online Content: Classical vs. Embedding-Based Approaches

Session 10 : AI for Healthcare & Clinical Applications

Chairs : Pr. S. NOUH, Pr. K. ACHTAICH, Pr. M. BOUTAIB, Pr. O. ZAKARY

S. CHAH, K. LI BOUCHTI, S. ZITI: A Narrative Review on Deep Learning Techniques for Breast Cancer Detection: Progress, Challenges, and Future Directions

F. ELGHANJAOU, F. ABOULKACIM: Intelligent System for Heart Disease Prediction

Z. ID-SAID, M. KASBOUYA, E. LOTFI: Deep Learning for Fractional Parameter Identification in Epidemic Models

A. LAARIBI: Optimal Treatment Strategy for an Epidemic Model with Reinfection and Relapse

Session 11 : Data Extraction, MLOps & AI Agents

Chairs : Pr. S. ELFILALI, Pr. S. OUNACER, Pr. O. ZAHOUR, Pr. B. AHARMIM, Pr. E. LOTFI

M. OUALI, K. EL BOUCHETI, L. ZRIMI, M. CHIBI, S. ZITI: Design of an Intelligent Agent for Proactive Prediction of Extreme Volatility Episodes in Financial Markets Using Noisy and Heterogeneous Data

L. ZRIMI, K. EL BOUCHTI, M. OUALI, M. CHIBI, S. ZITI: Intelligent Agents and Market Trend Reversal Prediction A Narrative Study

A. CHAHID, R. AIT ABDELOUAHID: Development of an AI Agent for Text-Based Semantic Image Search

CHAIMAA BOUAINA, CHAIMAE ZAOUI, FAOUZIA BENABBOU : Predicting Real Estate Prices through Machine Learning Approaches

S. OUJAA, A. BELANGOUR: Advances in Artificial Intelligence for Software Engineering: A Comprehensive Review

13H00-15H00

Lunch Break

15H00 – 17H00

Parallel Sessions (Block 4)



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Session 12 : AI Innovations Cybersecurity, Healthcare, and Human-AI Collaboration

Chairs : Pr. L. MOULAD, Pr. Z. ELLAKY, Pr. M. TAYEBI

- A. ZOUIR:** Towards Self-Adaptive Cybersecurity: Designing for Threat Detection, Prevention, and collective Defense Autonomous AI Agents
L. MOULAD, O. AIT ELMOUDDEN, M. KHIATI: Hybrid Intrusion Detection Model based on Support Vector Machine and Attack Signatures in Cluster-Based Sensor Network.
M. MAROUFI: Leveraging AI and Blockchain for Secure Healthcare Data Management: A Narrative Literature Review
S. AMAZIGH, S. ZITI: Mapping AI-enabled Cyber-physical Systems and IoT Research Landscape: A Bibliometric View
A. GHARBAOUI, A. ETTAOUFIK: Integrating Blockchain and Artificial Intelligence for Secure Educational Systems: A Systematic Comparative Review
S. FADAL-ELLAH: Human-AI collaboration
Z. BOUGUETAB: Cognitive Foundations for AI in Active Learning: A Critical Review

Session 13 : Applied AI Innovations: Recommender Systems, Multimodal Processing, and Explainable Decision-Making

Chairs : Pr. L. MOUSSAID, Pr. D. BOUGAR, Pr. H. FERJOUCHIA, Pr. K. BOUSKOURA

- A. ZAHID:** A Generative Text-Based Recommender System Leveraging Transformer Embeddings and Retrieval-Augmented Generation
E. LGHAOUCH, S. OUNACER, M. AZZOUAZI: Modeling Intra-Review Sentiment Divergence for Enhanced Recommendation Reliability
A. BOGRYN: Agentic AI-Enhanced MLOps Infrastructure for Automated Annotation and Training of Legal Machine Learning Models in Kubernetes Environments
T. AMZIL, A. JANNANI, A. BELHAOUS, N. SAEL: Development of a Multimodal Processing System for Speech and Emotion Recognition in Moroccan Arabic
A. LAAMOURI, N. SAEL: Visual Clothing Recommendation Using Vision-Language Models: A Comparative Study on Fashion-MNIST
A. ELMANSORI, A. EL MHOUTI, M. FAHIM: Explainability in Fake News Detection: The Role of SHAP and LIME
O. DEHBANE: Prediction to Performance: Trends & Challenges in Elite Sports Technology

Session 14 : Intelligent and Complex Systems

Chairs : Pr. R. AITABDELOUAHID, Pr. H. OUCHRA, Pr. K. KANDALI, Dr. N. SAFEH

- L. EL JIANI, S. EL Filali, E. BENLAHMAR, Z. BANOU, F. ALAOUI:** A Bibliometric Analysis of Large Language Models, RAG, and Agentic AI in Breast Cancer Imaging (2022–2025)
A. CHEBLI ALAMI, A. LOUDYI LAMGHAFRI, S. ZITI: AI in Media: A Narrative Review of Tools for Creating Audio and Visual Content
O. ELYAZYD, S. CHTITA, M. AIT DAOUD: Smart Computational Pipelines for Virtual Screening:



From Classical Docking to AI-Based Prediction Models

Z. NASSR, F. BENABBOU, N. SAEL : Rethinking Preprocessing in Moroccan Dialect Sentiment Analysis: Data Balancing Outperforms Traditional Stemming and Stop Word Removal

S. QAQA, F. BENABBOU, Z. ELLAKY: State-of-the-Art Review on AI-Driven Detection of Malicious Behaviors in Online Social Networks

M.Y. RACHIH, E. BENLAHMAR: Toward Efficient Text-to-Video Models under Limited Data: Transfer Learning and Rapid Adaptation Approaches.

15H00 – 17H00

Parallel Online Sessions (Block 2)

Session 3 : https://meet.jit.si/ICAIS25_session3

Chairs : Pr. M. HANOUNE, Pr. K. ELGUEMMAT, Pr. F. ELAZZABY, Pr. F.E. FAGROUD

Y. TAJANI, N. SAEL: Optimizing Fraud Detection on Imbalanced Data: A Comparative Study TPE and GridSearch using SMOTE

A. BAHASSOU, S. ZITI: From Graph Theory and Artificial Intelligence to Models of Economic Networks

H. SAKHI, S. EL FILALI, M. AMEUR , Y. ABOUDRAR: Rethinking Model Choice in Arabic Sentiment Analysis: A Comparative Study of Conventional and Pretrained Paradigms

Y. ELBAH: Artificial Intelligence and Digital Transformation in Public Administration: Towards Sustainable and Efficient Governance

L. IDOUGLID, S. TKATEK, K. ELFAYQ, A. GUEZ: Empowering IIoT Security with Deep Learning: A Feature-Driven Approach to Intrusion Detection Systems

A. OUCHANI, T. ZINAOUI: Digital Transformation and Artificial Intelligence in Moroccan Higher Education: Between Innovation, Reform, and Governance at Hassan II University of Casablanca

Session 4 : https://meet.jit.si/ICAIS25_session4

Chairs : Pr. S. OUHABI, Pr. A. ZAOUCH, Pr. D. BOUGGAR, Pr. N. AHARRANE

M. A. CHOURAK, Y. ZAOUI, D. RIANE, S. ZITI: AI-Driven Cyber Risk Exposure Management

Z. BANOUE, F. ALAOUI, S. EL FILALI, E. BENLAHMAR, L. ELJIANI: Computational and Cognitive Approaches to Qur'anic Linguistics: A Corpus-Based Perspective

A. ANANOUECH, H. KHALIFI, F. OUARDI: Federated Learning Meets Information Retrieval: A Survey of Recent Advances and Emerging Challenges

K. AANIBRA, A. BELANGOUR: Understanding the Challenge of Partial Object Recognition in Deep Learning

M. BAICH, N. SAEL: A Decentralized Deep Learning Approach to Intrusion Detection in IIoT Networks

Z. BANTISSI, F. BENABBOU: Toward Hybrid Reasoning in Healthcare Chatbots: Bridging Large Language Models, Biomedical Knowledge, and Clinical Applications

Y. ZNAGUI, S. OUAHABI, K. EL GUEMMAT: Towards Secure Healthcare Cloud Computing:



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Protecting Patients' Personal Data

Session 5: https://meet.jit.si/ICAIS25_session5

Chairs: Pr. A. ETTOUFIK, Pr. M. SADIK, Pr. S. OUNACER, Pr. T. AHAJJAM, Pr. M. GHAZOUANI

H. EL MORABIT, S. ZITI, W. RHALEM: 3D Semantic Segmentation Using Neural Networks and Connectivity Graphs for the Automated Extraction of Critical Anatomical Structures

Y. ASSINI, W. ABBAOUI, N. KHARMOUM, S. ZITI : Marché décentralisé et intelligent de l'énergie verte : vers une économie basée sur l'IA et la blockchain.

I. LAAROUA, M. OUZZIIF, K. BOURAGBA: Conceptual Framework for Edge-Based Federated Learning in Healthcare IoT Systems

Z. KEMAL, M. HANOUNE : Juges et modèles de régression : vers une évaluation par machine learning de la cohérence judiciaire

H. KBIRI ALAOUI, O. HAMAL, M. BENKHALIFA: Human-AI Teaming for Smarter, Safer Cities

K. SABIRI, E. GONÇALVES, L. AFONSO, R. FERNANDE: A Scalable VR-Based Training System for Workforce Upskilling in Human-Robot Collaborative Construction

A. IDSAID: AI and Industrial Property: How IP Systems Boost the Artificial Intelligence Revolution

15H00-17H00

Posters Session

Chairs: Pr. T. AHAJJAM, Pr. H. BOUTAYEB, Dr. O. AITLMOUDDEN, Dr. Y. MATRANE

M.A. FAHSSI: A New Approach to Image Encryption: Hyper-Quantum Fusion Combined with a Convolutional Neural Network.

M. ZIZOUANE: Topic Modeling for Moroccan Darija through Large Language Models: A Comparative Study

M. FADEL BENICHE, S. MOUNIRI : Développement d'un Assistant IA Contextuel pour l'Éducation : Exploitation des Graphes de Connaissances, RAG et Fine-Tuning

Y. TAKI, E. BENLAHMAR: Comparative Study of GAN, Transformer, and Diffusion Architectures for the Reconstruction of Historical Monuments

E. ELKAISSOUNI, E. BENLAHMAR: Autonomous Digital Twin for Precision Agriculture

A. ESSADIKINE: Implementation of a Visual Decision-Making System for the CNSS

M. RACHIH, E. BENLAHMAR: Towards Efficient Text-to-Video Models under Limited Data: Transfer Learning and Rapid Adaptation Approaches

Z. EL BIKARI, K. ACHTAICH, F.Z. OUAHOUDA: The Emotional Impact of Introducing Algorithmic Thinking in Primary School Children

F.Z. HOUSSA : Amélioration de la transcription Darija

A. KARROUM : Développement d'une application mobile de gestion et la localisation des équipements électriques intégrée à SAP

A. BAHAFID, A. BELANGOUR: Dashboard Generation

N. HAJJAOUI, A. BELANGOUR: Development of an AI-Based Mobile Application for English



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Learning with Conversational Features

N. ELBAH: Prédiction de la qualité de l'eau avec des modèles de machine learning

Y. NAJJAT: Transformation Automatique d'un Livre Arabe en un Dataset Structuré (IA + NLP)

A. MOSSATI, M. AIT DAOUD: Automated Scrum Assistant

A. ED-DAIF: Intelligent Forage Recommendation and Nutritional Optimization for Livestock Using AutoML and Advanced Feature Engineering

A. BRINI: Intelligent Conceptual Similarity Search in Arabic Texts

A. EL BOUSSANNI, M. AIT DAOUD: Math Tutor – An Adaptive Multi-Agent Intelligent System for Mathematics Learning

E. CHTIBA: Automated Extraction of Product and Quantity Fields from Purchase Orders Using Fine-Tuned Qwen2.5-VL-7B-Instruct

A. WAHBI, N. AHARRANE: Portfolio Optimization with Deep Reinforcement Learning

A. MISBAH, E. BENLAHMAR, Z. ELFAKIR, O. KAICH: Development of a model for detecting errors after data extraction

A. CHAHID: Development of an AI Agent for Text-Based Semantic Image Search

S. SALAMA: Creation of an Intelligent Platform for Connecting Freelancers and Companies with AI-Based Automatic Matching (OpportMatch)

R. ALASELY: Bearing Fault Detection Using the RMS Value of the Vibration Signal

A. SASSAOU, F. BENABBOU, K. ACHTAICH, M. AIT DAOUD: Early Payment Default Detection in the Leasing Sector Using Machine Learning: Moroccan use case

17H00-17H30

Coffee Break

17H30-18H00

Round table and Official closing



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